

2023 AMC 10A Problem 4 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 4

Problem:

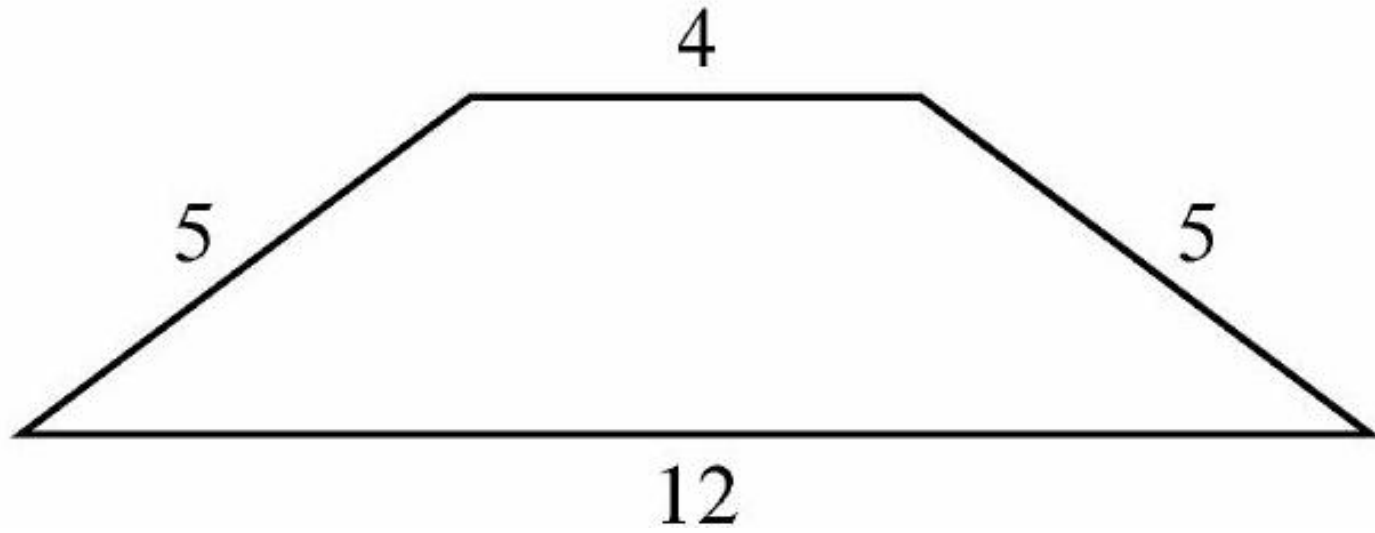
A quadrilateral has all integer side lengths, a perimeter of 26 , and one side of length 4. What is the greatest possible length of one side of this quadrilateral?

Answer Choices:

- A. 9
- B. 10
- C. 11
- D. 12
- E. 13

Solution:

The longest side must have length less than the sum of the other three side lengths. Because the sum of all the side lengths is 26 , this implies that the longest side length must be less than 13 , and because all the side lengths are positive integers, it must be less than or equal to 12 . Such a quadrilateral could have side lengths 12, 4, 5, and 5 , for example, as shown in the figure.



The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) [↗](#)

Powered by [Wiki.js](#)